

LASIK interface complications

Part 2 | Interface haze has competing mechanisms

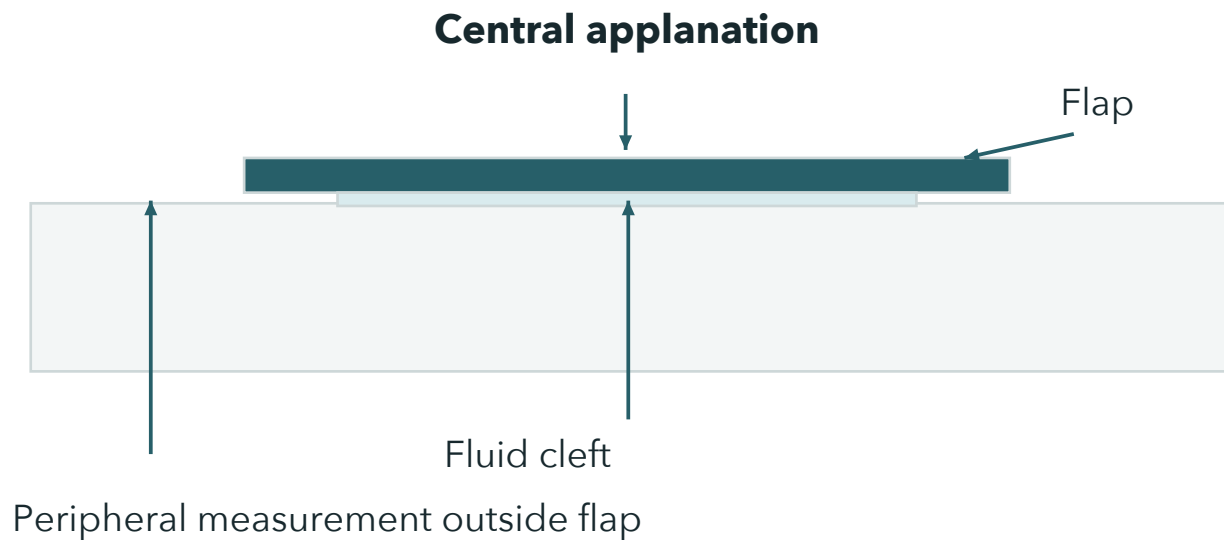
Timing, pressure and morphology guide the next action.

Pattern	Mechanism to investigate
Early diffuse granular interface cells	DLK after infection assessment
Later blur during steroid treatment	Pressure-related interface fluid
Epithelial tongues from flap edge	Epithelial ingrowth
Focal progressive infiltrate	Microbial infection

Flap interfaces can conceal both sterile and infectious disease. Document distribution, epithelial integrity, IOP and OCT findings. Timing supports but does not prove a diagnosis. Escalating steroid for every cloudy interface can worsen infection or pressure-induced disease, while premature steroid withdrawal can harm true severe DLK.

A low central reading can conceal high pressure

Measure away from the fluid-bearing interface when findings conflict.



Original section: the central flap deforms over fluid and can produce falsely low applanation.

Pressure-induced interface fluid may occur during steroid exposure. Central Goldmann readings can be misleading, so examine peripheral cornea outside the flap and use complementary assessment. Reduce the steroid burden and treat pressure when the mechanism is confirmed, with appropriate monitoring. Corneal thickness formulas do not correct a fluid-pocket artefact.

DLK needs a severity-aware response

Central aggregation and tissue change matter more than a generic haze label.

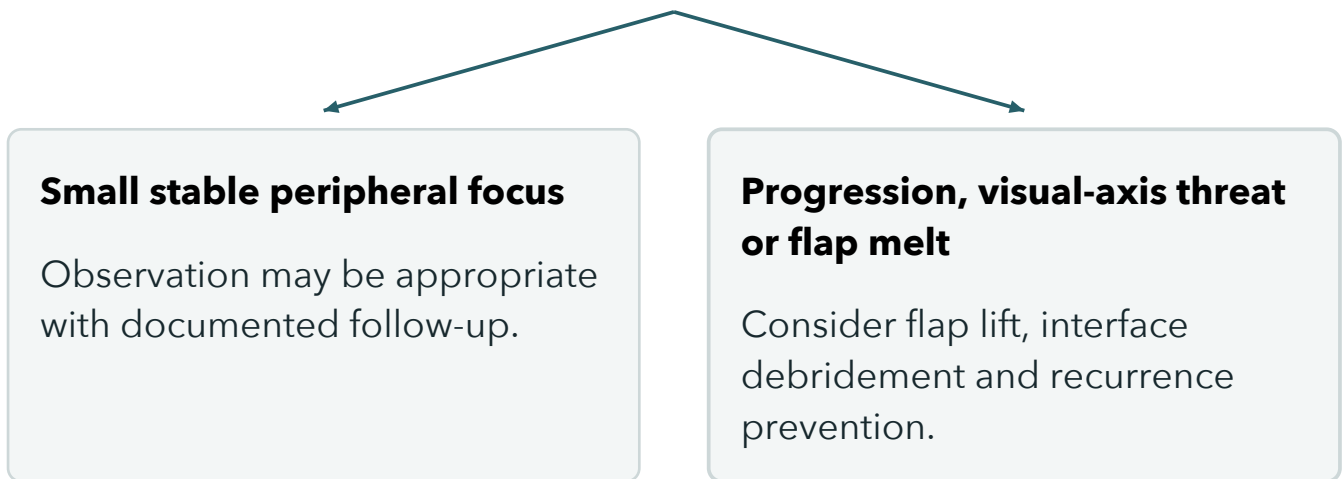
Observation	Implication
Peripheral granular inflammation	Monitor distribution and response closely
Central aggregation	Visual-axis involvement raises urgency
Stromal tissue loss	Permanent optical harm can follow
Atypical focal infiltrate	Reconsider infection before steroid escalation

DLK is sterile inflammation within the flap interface, often beginning early after surgery. Specialist treatment commonly uses corticosteroid, with flap-interface irrigation considered in significant disease. The plan depends on stage and trajectory. A photograph of haze alone cannot decide between DLK and infection.

Ingrowth is judged by progression and threat

Stable peripheral cells and advancing central disease are not equivalent.

What is the epithelial ingrowth doing?



Inspect the flap edge and compare serial extent. Recurrent disease may require measures that prevent re-entry, rather than repeated removal alone. An irreversible intervention should match the actual threat; a stable peripheral finding does not automatically require surgery, but progressive melt cannot be managed as an incidental observation.

Late ectasia is a different interface-era problem

Look for progressive shape change rather than treating haze alone.

Measure	What it contributes
Refraction and acuity	Functional change
Anterior and posterior shape	Ectatic geometry
Pachymetric pattern	Thinning distribution
Repeatability and lens history	Exclude artefact and warpage

Post-LASIK ectasia requires a progression and rehabilitation assessment, not another flap-inflammatory regimen. Evaluate prior surgery, tomography and competing causes of refractive change. Contact-lens rehabilitation and disease stabilisation can be complementary. A single reassuring anterior topography map does not fully assess the posterior cornea.